

GCD Subsets

Time limit: 1.00 s

Memory limit: 512 MB

You are given an array of n integers. Your task is to calculate the number of non-empty subsets whose elements' greatest common divisor is equal to k for each $k = 1, \dots, n$.

Input

The first line has an integer n : the size of the array.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print n integers as specified above modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq n$

Example

Input	Output
5 5 4 4 2 3	22 4 1 3 1